

Capstone Week 4 Project Plan + Gantt Chart

Capstone Project Plan + Gantt Chart — Week 4 Deliverable

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1. Working title (revised from Week 1-2 proposal)

Calibration Discipline as the Audit Primitive for Spatial Pre-Trade Compliance in Live Trading

(The Week 1-2 thesis — “*From Paper to Production: Calibration Discipline in a Solo-Operated AI Trading Agent*” — is preserved as the **Phase 1 core**. The extension proposed in the advisor amendment email of 2026-06-17 adds a measurement-first **Phase 2 spatial enforcement layer** that plugs into the validated Phase 1 backend without contaminating the Week 4-6 paper-trading validation window.)

2. Hypothesis and falsifiable predictions

H1 (core, unchanged). A multi-source AI agent can hold a measurable Brier-score advantage over a market-implied baseline when deployed live with small real capital (\$300-\$5k), across two heterogeneous asset classes — *iff* live paper P&L tracks backtest for ≥ 14 days before any real money.

H2 (added). A calibrated probability channel produced by Guo et al. 2017 temperature scaling reduces Expected Calibration Error by $\geq 50\%$ on a chronological held-out fold without degrading the agent’s rank-ordering (since temperature is monotone).

H3 (added). A Kadavath et al. 2022 second-order self-calibration gate, fired before staking capital, demotes BUY/SELL $\rightarrow$ WAIT at a rate that itself becomes calibration evidence: if the LLM’s own second-order confidence is well-calibrated, demote-rate at threshold 0.6 should correlate negatively with the resulting Brier of any decision that survives the gate.

3. Empirical strategy

Two test surfaces:

- **Polymarket binary events.** Baseline: market midpoint at the time of forecast. Brier surface is dense and resolves in days.
- **SpaceX-sector equities.** Eight pure-play tickers (RKL, ASTS, LUNR, BKS, PL, RDW, LMT, LIN). Baseline: buy-and-hold basket.

Per-prediction Brier $(p - o)^2$. Deliverable quantity: `Brier_agent - Brier_baseline` over rolling 30-day windows.

Public pre-registration: every prediction timestamped on `github.com/alex-jb/spacex-ipo-tracker` before it resolves → no cherry-picking.

Hard risk controls (binding throughout the live window): - 3-loss kill switch - \$300 deployment cap (no leverage) - 14-day ATR-scaled stops (2%–15%) - Sector exposure $\leq 30\%$ - **NEW:** confidence-shrunk Kelly (Phase 1 #5) — Kelly fraction $\times (1 - ECE/0.25)$, capped at quarter Kelly - **NEW:** PR-gated self-modification (Phase 1 #6) — no prompt/threshold edit merges to `main` without ≥ 0.005 Brier improvement on a 30+ held-out fold AND a human-approval flag

4. Six-week Gantt (Phase 1) + four-week extension (Phase 2)

WEEK	dates	calibration phase 1	spatial phase 2
1 (J2)	06-02	proposal interest statement	
2 (J9)	06-09	draft proposal (Brier thesis)	
3	06-16	annotated bibliography	
3.5	06-17 ●	★ Phase 1 #1-#6 modules shipped	(capstone
	amendment email	(ECE + temperature + self-cal	sent to advisor)
		+ reflexion + staleness +	
		Kelly shrink + D-G gate)	
		★ ece_audit.py LIVE: T* = 5.47	
		★ sc_audit.py LIVE: 100% dem	
4	06-23	weekly_brier_report extends	
		to ECE + SC dashboards	
		reflexion JSONL wired into	
		nightly brier_audit	
5	06-30	confidence-shrunk Kelly in	
		paper-trading (not real \$)	
		first synthetic D-G self-mod	
		proposal → first MergeDecision	
		recorded	
6	07-07 ●	midterm presentation:	
		ECE before/after, SC demote	

		rate, Brier delta, Phase 1	
		results writeup	
			▼
7	07-14		Quest 3S WebXR
scene +			
layout			Three.js podium
setup			MediaPipe hands
8	07-21		crypto hash chain
audit			trail (WebCrypto)
			first gesture-
vote loop			
9	07-28		spatial layer
integrated			
			with calibrated
backend,			
			end-to-end audit
export			
10 (●)	08-05	final presentation + reflection paper	
17	08-12	final project write-up due	

★ = items already done as of 2026-06-17 (this submission). ● = course-marked milestones.

5. Phase 1 status as of this submission

#	Upgrade	Source	Module	Tests	Live?
1	ECE + temperature scaling	Guo 2017 (ICML)	src/calibration/ece.py , temperature.py , reliability.py	19 ✓	Yes (ece_auc)
2	LLM self-calibration	Kadavath 2022	src/calibration/self_calibration.py	10 ✓	Yes (sc_audit)
3	Structured Reflexion JSON	Shinn 2023 (NeurIPS)	src/calibration/reflexion.py	8 ✓	Phase 1 wiring W 4
4	Information staleness	O'Hara 1995	src/calibration/staleness.py	10 ✓	Phase 1 wiring W 4
5	Confidence-shrunk Kelly	Kelly 1956 + Guo 2017	src/calibration/kelly.py	9 ✓	Paper tr: Week 5

#	Upgrade	Source	Module	Tests	Live?
6	PR-gated Darwin- Gödel	Sakana 2025	src/calibration/self_modification.py	8 ✓	First synthetic Week 5

Code: 64 unit tests green, 0.3s.

6. Two live findings already on the table (Week 3.5)

Finding F1 — Systematic over-confidence pathology reproduces in production

scripts/ece_audit.py on decision_log.json (n=279 resolved, lookahead=1 day):

- ECE (raw): 0.0459
- Brier (raw): 0.2557
- **Fitted temperature T^* on chronological held-out fold (newest 25%): 5.474**
- ECE improvement on validation fold: +0.0494 (cuts ECE by roughly half)
- Brier improvement: +0.0056

Guo et al. (2017) showed that modern deep networks are systematically over-confident, with calibrated temperatures typically in [1.5, 3.0]. Our $T^* = 5.474$ indicates a *more severe* over-confidence pathology in LLM-derived trading probabilities than Guo's CNN benchmarks. This is a finding worth defending.

Finding F2 — LLM second-order self-knowledge would have demoted every recent directional call

scripts/sc_audit.py on last 24h BUY/SELL decisions (n=8) calling Claude Haiku 4.5 with the Kadavath-style second-order prompt:

- Average second-order confidence: 0.338 (well below the 0.60 demote threshold)
- **Demote rate at threshold 0.60: 100%**
- Top primary-doubt cluster: "57% barely exceeds random chance; lacks fresh data and clear regime"

Kadavath et al. (2022) argued language models (*mostly*) *know what they know*. F2 reproduces that finding **at the trading-agent level**: the same model family used to *produce* the directional probability, when asked a second-order question, says every recent directional call was too weakly supported to merit conviction. This is the inner loop of the falsification gate.

F1 + F2 are converging signals that the raw probability stream is over-stated, and they suggest two independent recalibration channels (temperature scaling + self-calibration gating) that are themselves testable hypotheses for Phase 1.

7. Risk register

Risk	Likelihood	Impact	Mitigation
14-day paper validation fails again, ATR fix not enough	Medium	High	iff gate triggers; do not deploy real \$ until reproduces. Document and analyze. Honest negative is publishable.
Phase 2 spatial layer slips past Week 10	Medium	Medium	Phase 1 alone is the defensible capstone claim; Phase 2 is “future work” if not done by Aug 12.
yfinance throttling delays nightly audits	High	Low	Existing brier_audit has retry logic; ece_audit reuses it. Add caching layer if needed.
Claude API cost overruns from sc_audit	Low	Low	Haiku 4.5 batch is cheap (~\$0.0005 per audit call); rate-limited; daily cron caps at 50 calls/day = ~\$0.025/day
Quest 3S hardware not borrowed/purchased by Week 7	Medium	High (Phase 2)	Phase 2 can be demoed in browser fallback (WebXR + desktop view) without headset for thesis defense
Yang declines the scope amendment	Low	Medium	Revert to Phase 1-only thesis (no spatial layer). The 6 calibration upgrades alone constitute a substantial capstone.

8. Conference paper plan (out-of-scope for capstone defense, in-scope for combined thesis)

Target: ACM ICAIF 2027 (estimated submission deadline 2026-09-15) **Backup:** IEEE BigData 2026 (December submission) **Title (working):** *Embodied Calibration: Spatial Pre-Trade Compliance Grounded in a Brier- and ECE-Audited Multi-Source Agent* **Co-author plan:** Sole-author capstone deliverable; conference paper invites co-authorship from collaborators on the spatial extension.

9. What I'd value from the advisor in Week 4

1. A gut-check on the scope amendment: does the Phase 1 → Phase 2 framing read as natural sharpening, or scope creep I should pull back from?
 2. Recommendations on out-of-sample validation literature beyond Tetlock and Christoffersen for the thesis introduction.
 3. A formal advisor signature for capstone credit if pending.
 4. Any objection to public-by-default repos (the capstone is in `github.com/alex-jb/capstone-orallexa-calibration`, MIT-licensed, no real-money trade logs committed) — the public timestamping is part of the methodology, but I want to make sure you're comfortable with the visibility.
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10. Files in this submission

- `proposal/week-4-project-plan-and-gantt.md` (this document)
- `proposal/proposal.md` and `proposal.pdf` (Week 2 baseline; unchanged)
- `proposal/references.bib` (annotated bibliography source)
- `ANNOTATED-BIBLIOGRAPHY.md` and `Annotated-Bibliography.pdf` (Week 3 deliverable)
- `Advisor-Meeting-Brief-Yang.md` and `.pdf` (Week 3 verbal)
- `src/calibration/` (Phase 1 #1-#6 implementations + tests)
- `scripts/ece_audit.py`, `scripts/sc_audit.py` (live audits)
- `data-snapshots/ece-2026-06-17.{json,md}` (F1 evidence)
- `data-snapshots/sc-2026-06-17.{json,md}` (F2 evidence)
- `weekly-reports/` (Week 1, Week 2 retros)